

Appendix

1 Normalized score

Scaling performance using expert performance is a common evaluation strategy in RL [1, 2]. In our problem, we define the reward as the negative distance:

$$r_t = -r(p_t^{obj}) = -\|p_t^{obj} - p_g^{obj}\|_2. \quad (1)$$

For a given goal p_g^{obj} , $r_t \in]-\infty, 0]$. In order to compare different tasks, where distances may have different magnitudes, we divide the rewards r_t by the typical reward range. This is given by $r_{max} - r_{min}$, where $r_{min} = r(p_0^{obj})$, with p_0 being the initial position of the object (this is normally around the origin, and $r_{max} = r(p_g^{obj}) = 0$).

Thus, we obtain:

$$\text{score}_t = r_t / (r_{max} - r_{min}) \quad (2)$$

$$= r(p_t^{obj}) / (0 - r(p_0^{obj})) = \quad (3)$$

$$= -\|p_t^{obj} - p_g^{obj}\|_2 / (0 + \|0 - p_g^{obj}\|_2) \quad (4)$$

$$= -\|p_t^{obj} - p_g^{obj}\|_2 / \|p_g^{obj}\|_2 \quad (5)$$

Finally, we apply the exp operator, to make values positive and bring them in the $[0, 1]$ range, where 1 is the expert score:

$$\text{normalized score} = \exp\left(-\frac{\|p_t^{obj} - p_g^{obj}\|_2}{\|p_g^{obj}\|_2}\right) \quad (6)$$

2 FOCUS objective

Training of the FOCUS architecture is guided by the following loss function:

$$\mathcal{L}_{\text{FOCUS}} = \mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{state}} + \mathcal{L}_{\text{obj}}. \quad (7)$$

$\mathcal{L}_{\text{dyn}}$ refers to the dynamic component of the RSSM, and equals too:

$$\mathcal{L}_{\text{dyn}} = D_{\text{KL}}[p_\phi(s_{t+1}|s_t, a_t, e_{t+1}) || p_\phi(s_{t+1}|s_t, a_t)]. \quad (8)$$

the backpropagation is balanced and clipped below 1 nat as in DreamerV3 [5].

The object loss component is instantiated as the composition of NLL over the mask and RGB mask reconstructions:

$$\mathcal{L}_{\text{obj}} = -\log p(\hat{m}_t) - \log \sum_{\text{obj}=0}^N \underbrace{m_t^{\text{obj}} p_\theta(\hat{x}_t^{\text{obj}} | s_t^{\text{obj}})}_{\text{masked reconstruction}} \quad (9)$$

Finally, the decoder learns to reconstruct the rest of vector state information v_t by minimization of the negative log-likelihood (NLL) loss:

$$\mathcal{L}_{\text{state}} = -\log p_\theta(\hat{q}_t, p_g^{obj} | s_t) \quad (10)$$

3 Training details and Hyperparameters

All models in the presented work have been trained offline. Datasets have been collected beforehand, guided by the exploration agent of choice (we tested both the Object-centric entropy maximization proposed in FOCUS [3] and Plan2Explore [6]). The datasets are loaded in the replay buffer of the offline agents, and the training is conducted for 250K steps. Both world model and agent are updated at every training step. V100-16GB GPUs have been used for all experiments. Our proposed methods (i.e. Dreamer/FOCUS + PCP, FOCUS + LCP) took roughly 18 hours to complete each training run.

The hyperparameters used for the main implementation of the world models and agent are the same used in DreamerV2 [4] official implementation. Symlog function is applied at every input. KL balancing as in DreamerV3 [5] is implemented.

With reference to FOCUS model, we have the following additional parameters:

- Object-extractor: MLP composed of 2 layers, 512 units, ReLU activation;

With reference to FOCUS + LCP model, we have the following additional parameters:

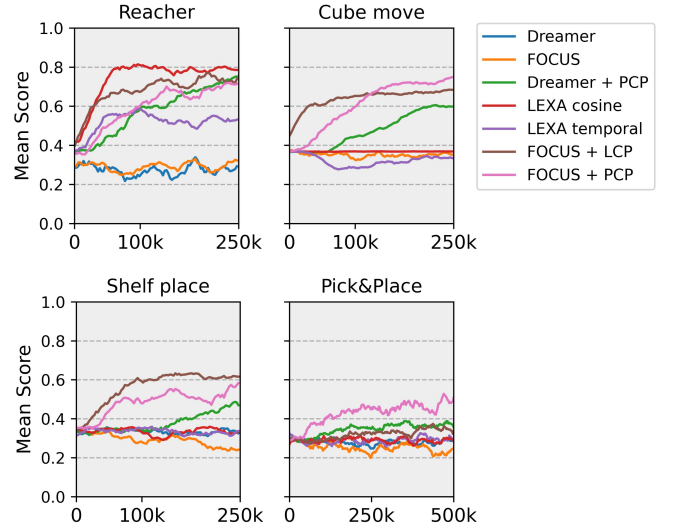


Figure 1. Offline training curves. Standard deviation is omitted for graphical reasons. Mean score refers to eq. 2 (main paper)

and is computed over 5 evaluation episodes, performed during the offline training. For each episode, a random goal is selected out of a pool of 10 manually engineered ones.

- Object-encoder: MLP composed of 4 layers, 400 units, ReLU activation;
- Distance method object-encoder objective: Cosine similarity (also tested MSE)
- Distance method actor policy objective: Cosine similarity (also tested MSE)

4 Offline Training Curves

Offline training curves are presented in Figure 1. In general FOCUS + PCP/LCP have faster convergence when compared to all other methods. Only for the Reacher environment, LEXA cosine converge faster.

5 Explorations strategies

In the presented work each model is trained offline from a pre-recorded dataset. The dataset of choice is obtained from pure exploration behavior. In Fig. 2 we compare the general performance of LCP when trained on datasets acquired using different exploration strategies. We consider the object-centric entropy maximization method proposed by Ferraro et al. [3] and Plan2Explore [6].

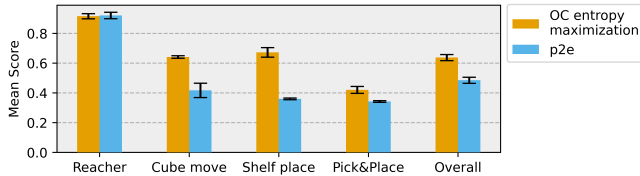


Figure 2. Mean score achieved over 10 episodes for models trained with both datasets obtained from FOCUS exploration method (Object-Centric entropy maximization) and Plan2Explore. The score is expressed according to equation 2 (main paper).

Overall exploring by maximizing the entropy over the object’s latent, gives better performance in the downstream task. We hypothesize this is related to the focus the exploration strategy puts on the object of interest while disregarding background aspects in the scene.

References

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